

Detecting Adverse Drug Events in Clinical Notes Using Large Language Models

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Abstract. Monitoring adverse drug events (ADEs) is critical for pharmacovigilance and patient safety. However, identifying ADEs remains challenging, as suspected or confirmed side effects are often documented solely in the unstructured text of electronic health records (EHRs). Manually reviewing clinical notes to detect ADEs is labor-intensive and time-consuming, highlighting the need for automated methods capable of analyzing and extracting ADE-related information from clinical documentation. In this short communication, we describe our ongoing research on fine-tuning and evaluating a large language model (LLM) for the detection of ADEs in clinical notes. Preliminary descriptive result of this study indicates that ADEs are poorly documented in discharge notes, with less than 15% explicitly linking ADEs to specific drugs, which highlights the need for improved reporting practices.

Keywords. Adverse drug event, discharge notes, medical named entities, large language model

1. Introduction

Medications are essential to healthcare but can also lead to adverse drug events (ADEs), which may be harmful or even fatal. Detecting ADEs poses significant challenges, as they are often underreported or not documented in a structured format within electronic health records (EHRs) [1]. Many ADEs are instead described in the unstructured text of clinical notes [2]. Over the years, various approaches have been explored to develop effective methods for ADE detection in clinical notes, ranging from early rule-based systems to machine learning-based methods [3], and more recently, large language models (LLMs) [4]. In our ongoing research, the LLM Swe-De-Clin-ADE-BERT [5] performs well in certain ADE detection tasks (with macro average F1 score 0.84 for medical named entity recognition (NER) task) but falls short in others (with F1 score 0.70 for ADE relation extraction (RE) task), indicating the need for additional data to

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further fine-tune the model. The aim of this study is to fine-tune and evaluate an LLM for the detection of ADEs in clinical notes.

2. Methods and Results

In this study, we utilized retrospective data from the electronic health records (EHR) of a Geriatric Clinic in a Swedish region, covering the period from 2018 to 2022. We selected 905 discharge notes with ADE diagnosis codes representing 199 patients. To detect ADEs in Swedish medical texts, we chose the Swedish BERT model trained on clinical texts (Swe-De-Clin-BERT). Unlike generative decoder-based models like OpenAI's or Llama, this model offers contextual relevance, task-specific training (e.g., NER, RE), domain and language focus, transparency, privacy compliance, and lower computational demands. The notes were initially pre-annotated with medical entities using the fine-tuned Swe-De-Clin-BERT model [5] and further refined entities and annotated the relations by two domain experts (pharmacists) using the BRAT tool [5]. This study was approved by the Swedish Ethical Review Authority (Dnr. 2023-06920-01) and funded by the Linnaeus University Center DISA.

Among the annotated text 56% included text describing the ADEs. Some texts where ADEs were mentioned included information about which drugs were believed to cause the ADE, and which findings were related to the ADE (such as *dizziness*, *confusion*, *nausea*, and *low blood pressure*). Only 14.9% (135) of the discharge notes has explicitly mentioned ADEs along with a specific drug name, showing the need for improved reporting.

3. Discussion and Conclusions

The next step will be to further fine-tuning the LLM on these annotated clinical notes to improve the ADEs detection accuracy and find ADEs in non-annotated discharge notes. Effective methods to detect ADEs documented in an unstructured form will be valuable in increasing knowledge to prevent ADEs and improve patient safety.

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